CLINICAL · MOUTH & THROAT

# Adenoids and nasopharyngitis

A child who breathes through the mouth, snores, has a constant wet nose and a dull, hearing-impaired look is the classic child with adenoids. Size alone does not matter; what matters is the adenoid relative to the nasopharynx it fills, and what it does to the nose, the ears and sleep. Most children outgrow the problem; those who do not, and whose ears or breathing suffer, are helped by a short operation.

**AT A GLANCE**

- Adenoids have no crypts and no capsule.
- Nasal, aural and general symptoms; adenoid facies.
- Submucous cleft is a contraindication to adenoidectomy.
- Grisel syndrome and atlantoaxial instability in Down syndrome.

## 1 CHAPTER ONE The adenoid

The nasopharyngeal tonsil.

### 1.1 Anatomy

The adenoid lies at the junction of the roof and posterior wall of the nasopharynx, as vertical ridges of lymphoid tissue separated by deep clefts. It is covered by ciliated columnar, stratified squamous and transitional epithelium and, unlike the palatine tonsil, has *no crypts and no capsule*.



**FIGURE 1** An adenoid mass after removal with a curette: ridges of lymphoid tissue separated by deep clefts.

Dhingra, fig. 48.1

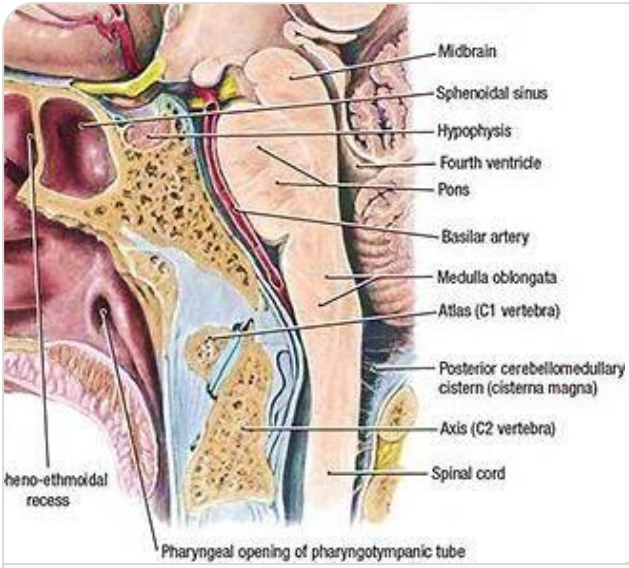



**TABLE 1** Differences between palatine tonsils and adenoids

| | Adenoids | Palatine tonsils |
|---|---|---|
| **Number** | Single | One on each side |
| **Site** | Nasopharynx | Tonsillar fossa in oropharynx |
| **Crypts or furrows** | Only furrows | Only crypts |
| **Capsule** | Absent | Present |
| **Epithelium** | Ciliated columnar | Squamous stratified |
| **In adults after 20 years of age** | Absent | Present |

**FIGURE 2** The nasopharynx in sagittal section, and the adenoids compared with the palatine tonsils.

Anatomy atlas · ENT Steps p. 200, ENT Steps p. 208

## 1.2 Growth and regression

Present at birth, it enlarges physiologically to a peak at about three to seven years, regresses from puberty and has largely gone by twenty; allergy or infection can

keep it large beyond fifteen.

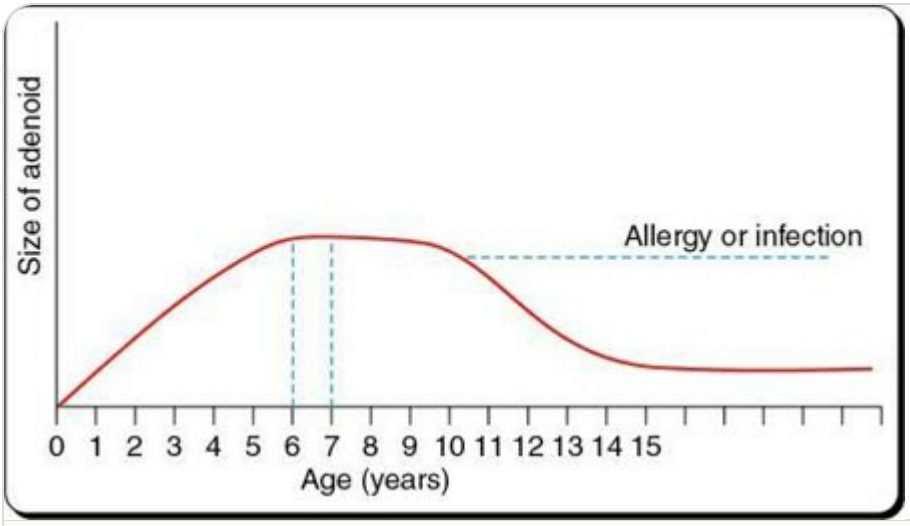



**FIGURE 3** Size of the adenoid with age: maximum at six to seven years, regressing at puberty; persistence beyond fifteen with allergy or infection.

Dhingra, fig. 48.2

### 1.3 Supply

- **Arteries:** ascending palatine (facial), ascending pharyngeal, the pharyngeal branch of the maxillary artery, and the ascending cervical branch of the inferior thyroid.
- **Lymph:** to the upper jugular nodes, directly or through the retropharyngeal and parapharyngeal nodes.
- **Nerves:** glossopharyngeal and vagus, which explains the earache of adenoiditis.

## 2 CHAPTER TWO Adenoid hypertrophy

Too big for the space.

### 2.1 Causes

Physiological enlargement in childhood, a general tendency to lymphoid hyperplasia, recurrent rhinitis, sinusitis and tonsillitis, and allergy.

## 2.2 Clinical features

### 2.2.1 Nasal

- **Obstruction:** the commonest symptom, with mouth breathing; infants struggle to feed because they cannot breathe and suck at once, and fail to thrive.
- **Discharge:** secretions cannot drain back through the blocked choanae, and chronic rhinitis adds to them: the "wet, bubbly nose".
- **Sinusitis:** chronic maxillary sinusitis and adenoid infection feed each other.
- **Epistaxis** when the adenoid is inflamed.
- **Voice:** hyponasal (*rhinolalia clausa*), losing the nasal consonants.

### 2.2.2 Aural

- **Eustachian tube obstruction:** a retracted drum and conductive deafness.
- **Recurrent acute otitis media** from spread of infection up the tube.
- **Otitis media with effusion:** the adenoid is an important cause; the fluctuating size gives a fluctuating hearing loss.
- **Chronic suppurative otitis media** that will not settle while the adenoid is infected.

### 2.2.3 General

- **Adenoid facies:** an elongated face with a dull expression, open mouth, prominent crowded upper teeth, short upper lip, pinched nose from disuse of the alae, and a high-arched palate because the tongue no longer moulds it.
- **Sleep-disordered breathing:** snoring and obstructive sleep apnoea, with daytime tiredness and poor concentration (*aprosexia*).
- **Pulmonary hypertension and cor pulmonale** with long-standing severe obstruction.



**FIGURE 4** Adenoid facies: the open mouth of the mouth breather.
Dhingra, fig. 48.3

**SIZE IS RELATIVE**

A large adenoid in a large nasopharynx may cause no symptoms; a modest one in a small nasopharynx can block it completely. Judge the airway, not the adenoid.

## 2.3 Diagnosis

- **Flexible nasendoscopy** in a cooperative child: the best view of the adenoid, the choanae and the tube openings, without radiation.
- **Lateral soft tissue radiograph** of the nasopharynx: shows the adenoid and the remaining airway; useful in children who will not tolerate a scope, though crying lifts the palate and can exaggerate the size.
- Mirror examination is possible in some older children.
- Examine the whole nose for other causes of obstruction, and the ears with tympanometry and audiometry.

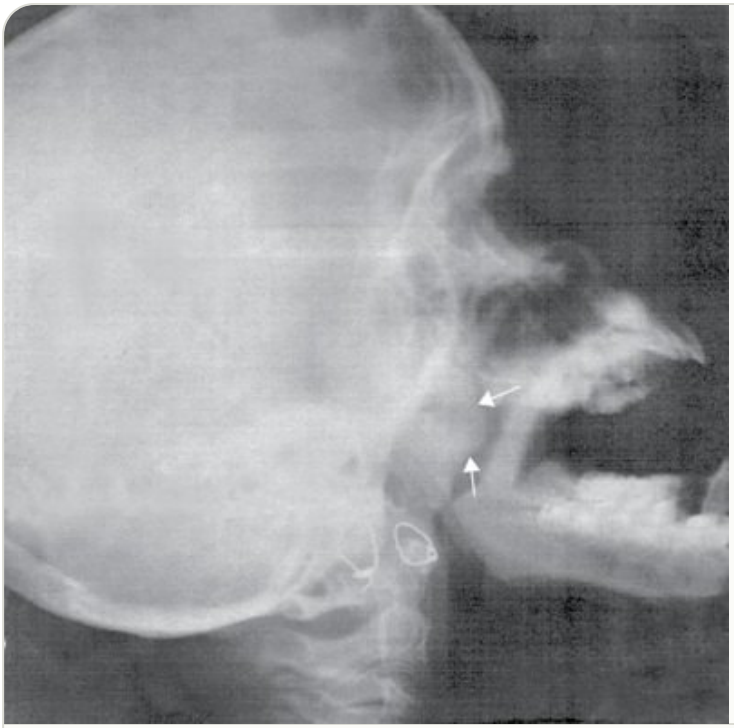


**FIGURE 5** Enlarged adenoids (arrows) on a lateral radiograph of a seven-year-old girl, leaving very little airway.

Dhingra, fig. 48.4

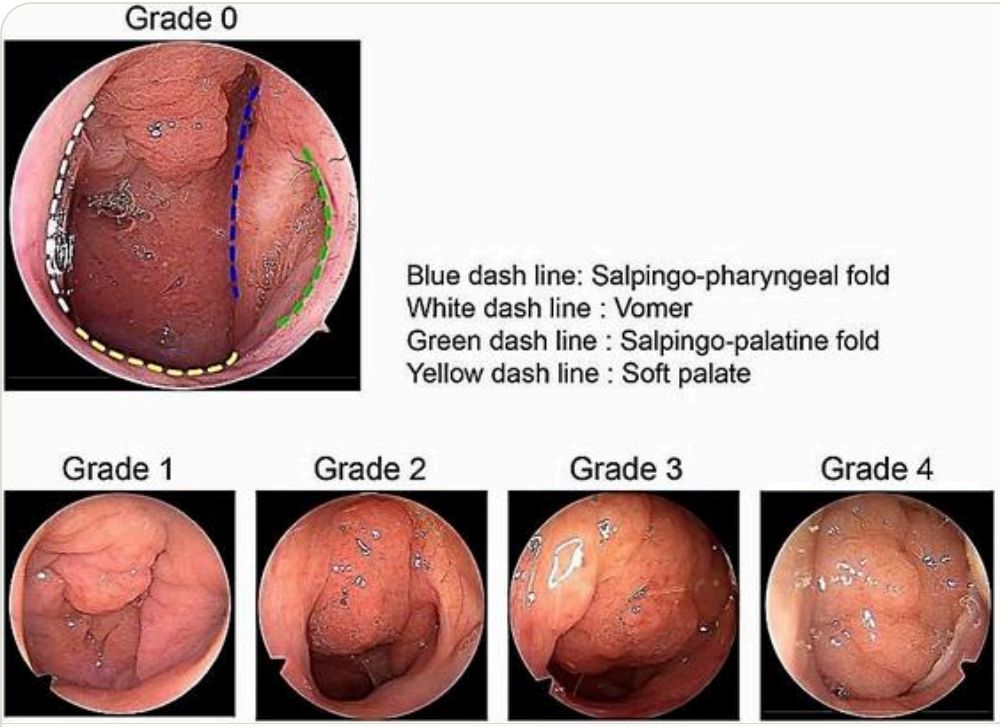
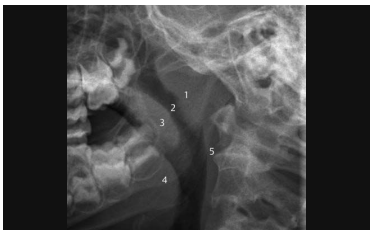



**FIGURE 6** Endoscopic grading of adenoids.

Anatomy atlas · ENT Steps p. 207

**IMAGING IN THE APP** 1 case

Adenoid hypertrophy on imaging.




**Adenoid Hypertrophy**

Infection & inflammation · 1 image

## 2.4 Treatment

### 2.4.1 Medical

For milder symptoms: intranasal steroid sprays, with montelukast in some children, treatment of allergy, saline and decongestant drops for short periods, and breathing exercises. Many children improve as the adenoid regresses.

### 2.4.2 Adenoidectomy

For marked symptoms. Dhingra's indications:

- Adenoid hypertrophy causing snoring, mouth breathing, sleep apnoea or hyponasal speech.
- Recurrent rhinosinusitis.
- Otitis media with effusion associated with adenoid hyperplasia (usually with grommets).
- Recurrent ear discharge in chronic otitis media associated with adenoiditis.
- Dental malocclusion: adenoidectomy does not correct it but helps prevent relapse after orthodontic treatment.

**CONTRAINDICATIONS**

- **Cleft palate or submucous cleft:** removing the adenoid that helps close the velopharynx causes velopharyngeal insufficiency. Look for a bifid uvula, a notch in the hard palate and a translucent midline zone (*zona pellucida*).
- **Bleeding disorders.**
- **Acute upper respiratory infection.**

# 3 CHAPTER THREE The operation

Adenoidectomy.

## 3.1 Technique

General anaesthesia with an oral tube; the same position as tonsillectomy, but never hyperextend the neck. With the Boyle–Davis gag in place, the nasopharynx is examined by retracting the palate and by palpation (feeling also for a submucous cleft). An adenoid curette with a guard is passed until it touches the back of the septum, pressed back and swept down, with the head slightly flexed to protect the atlas; lateral tags are removed with smaller curettes or punch forceps, taking care of the Eustachian tube openings. Bleeding is stopped by packing, with diathermy for persistent points; rarely a postnasal pack is left for 24 hours. When the tonsils are also to come out, the adenoid is removed first and the nasopharynx packed.

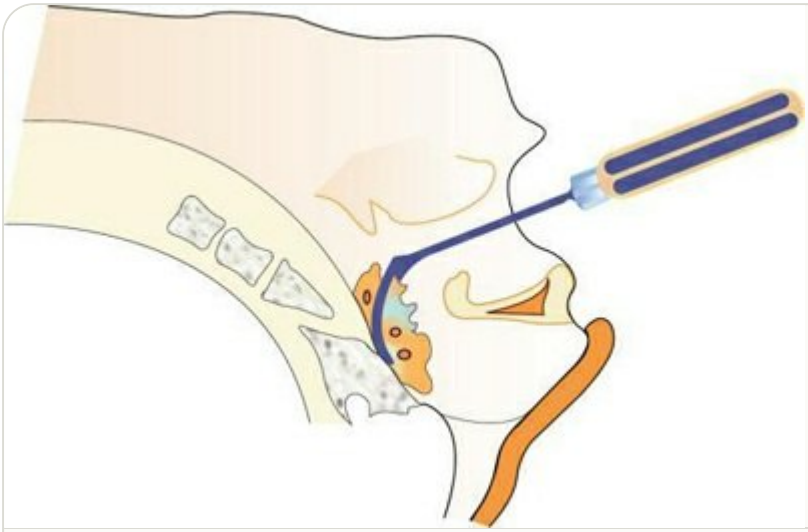


**FIGURE 7** Adenoidectomy with the curette.
Dhingra, fig. 95.1

## 3.2 Endoscopic techniques

Suction diathermy, a microdebrider or coblation under endoscopic or mirror vision remove the adenoid more completely, including tissue around the tube openings and choanae.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Adenoidectomy.

**Adenoidectomy**
Adenoidectomy

## 3.3 Complications

- **Haemorrhage:** usually primary; watch for a rising pulse and the vomiting of dark swallowed blood. A postnasal pack under general anaesthesia may be needed.
- **Injury to the Eustachian tube openings.**
- **Injury to the pharyngeal muscles and cervical spine** from hyperextension or pressure; 10 to 20% of children with Down syndrome have atlantoaxial instability.
- **Grisel syndrome:** neck pain and torticollis from inflammatory subluxation of the atlantoaxial joint, needing a collar or traction.
- **Velopharyngeal insufficiency:** hypernasal speech and nasal regurgitation, usually transient, but persistent with an unrecognised submucous cleft.
- **Nasopharyngeal stenosis** from scarring.
- **Regrowth** of tissue left behind; Lee advises biopsy of recurrent regrowth to exclude lymphoma.

# 4 CHAPTER FOUR Other nasopharyngeal inflammations

Nasopharyngitis and Thornwaldt disease.

## 4.1 Acute nasopharyngitis

Part of a common cold or influenza, or a bacterial infection (streptococcus, pneumococcus, *Haemophilus*). Dryness and burning behind the nose, then pain and difficulty in swallowing; in severe cases fever and nodes. The mucosa is red, swollen and covered in exudate. Most settle with analgesia; antibiotics for severe bacterial infection; decongestant drops for the associated adenoiditis of children.

## 4.2 Chronic nasopharyngitis

Adults with chronic nasal, sinus and pharyngeal infection, heavy smokers and drinkers, and those exposed to dust and fumes. Postnasal discharge and crusting, constant hawking and snorting back of secretions; congested mucosa with mucopus or crusts. Treat the source, stop smoking and drinking, and use alkaline douches and steam inhalation.

## 4.3 Thornwaldt disease

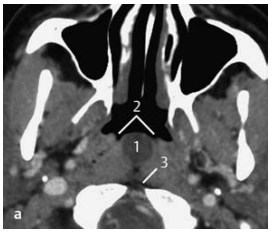
Infection of the *pharyngeal bursa*, the midline recess within the adenoid marking the embryonic attachment of the notochord.

- Persistent postnasal discharge with crusting.
- Nasal obstruction, Eustachian tube blockage and otitis media with effusion.
- A dull occipital headache, recurrent sore throat and low fever.

A cystic, fluctuant midline swelling sits on the posterior nasopharyngeal wall. Antibiotics, then marsupialisation and removal of the lining. Asymptomatic Thornwaldt cysts found on MRI need no treatment.

**IMAGING IN THE APP** 1 case

A Thornwaldt cyst.




**Thornwaldt Cyst**
Congenital · 2 images

**THE ADULT WITH A ONE-SIDED NASOPHARYNGEAL PROBLEM**

An adult with unilateral otitis media with effusion, a nasopharyngeal mass or a neck node is examined and, if needed, biopsied for nasopharyngeal carcinoma. Adenoid tissue in an adult is unusual; consider HIV and lymphoma.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Adenoid versus palatine tonsil:** no crypts, no capsule.
- **Symptoms** under nasal, aural and general headings; **adenoid facies.**
- **Lateral radiograph** and its limits.
- **Indications and contraindications** for adenoidectomy, especially submucous cleft.
- **Complications,** including Grisel syndrome and atlantoaxial instability in Down syndrome.
- **Thornwaldt disease.**

## 5.2 Common pitfalls

COMMON PITFALLS

- Removing the adenoid without feeling the palate for a submucous cleft.
- Hyperextending the neck of a child with Down syndrome.
- Calling a nasopharyngeal mass in an adult "adenoids" without a biopsy.

Written for this app from Oxford Textbook of Otolaryngology ch. 87, 88, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 10, Dhingra ch. 48, 95 and K. J. Lee ch. 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

## Adenoid Hypertrophy

### IMAGING FINDINGS

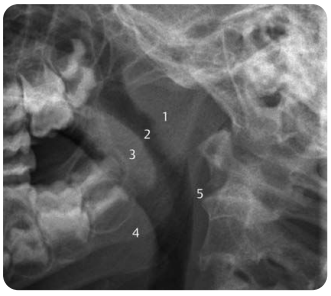
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

### TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

### CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions

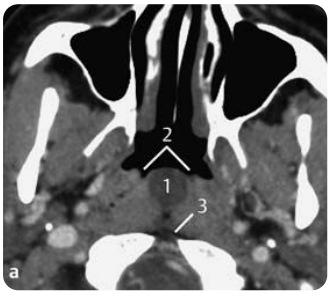



Plain radiograph for adenoid evaluation.

Dunnebier, Imaging for Otolaryngologists, Fig. 7.2

PHARYNX, TONSILS & OESOPHAGUS · CONGENITAL

## Thornwaldt Cyst




CT, axial. During endoscopy, a rounded expansion on the posterior wall of the nasopharynx in a patient with sinonasal symptoms was noted. CT…